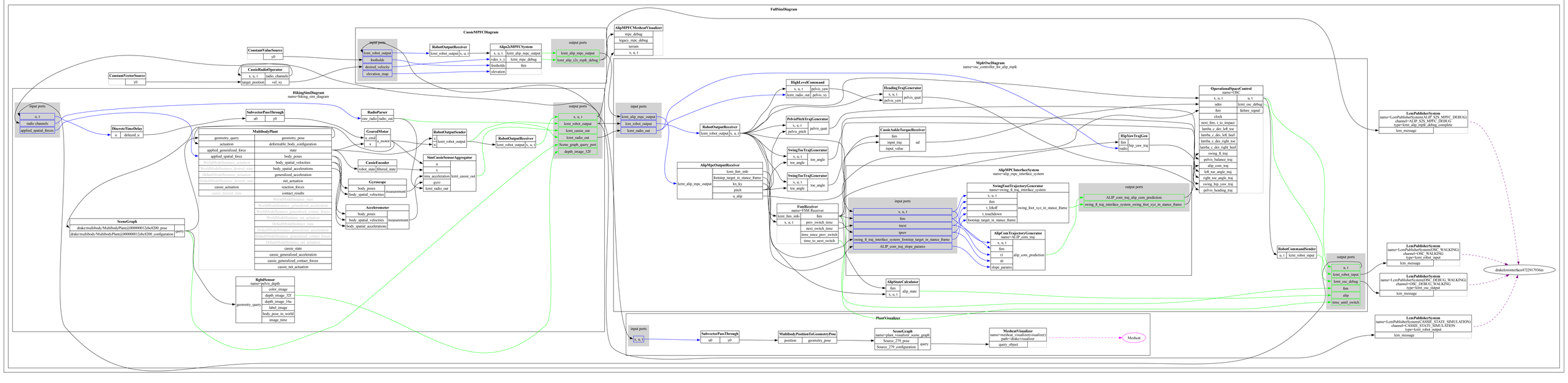
Diagram
FullSimDiagram
CassieMPFCDiagram
input ports
kcmt_robot_output
footholds
desired_velocity
elevation_map
output ports
kcmt_alip_mpc_output
kcmt_alip_s2s_mpfc_debug
RobotOutputReceiver
kcmt_robot_output
x, u, t
Alip2sMPFCSystem
x, u, t
kcmt_alip_mpc_output
vdes_x_y
kcmt_mpc_debug
footholds
fsm
elevation
AlipMPFCMeshcatVisualizer
mpc_debug
legacy_mpc_debug
terrain
x, u, t
ConstantValueSource
y0
ConstantVectorSource
y0
CassieRadioOperator
x, u, t
target_position
radio_channels
vel_xy
HikingSimDiagram
name=hiking_sim_diagram
input ports
x, u, t
radio channels
applied_spatial_forces
output ports
x, u, t
kcmt_robot_output
kcmt_cassie_out
kcmt_radio_out
Scene_graph_query_port
depth_image_32f
DiscreteTimeDelay
u
delayed_u
SubvectorPassThrough
u0
y0
RadioParser
raw_radio
radio_out
GearedMotor
u_cmd
x
u_motor
RobotOutputSender
x
u
kcmt_robot_output
RobotOutputReceiver
kcmt_robot_output
x, u, t
MultibodyPlant
geometry_query
actuation
applied_generalized_force
applied_spatial_force
WorldModelInstance_actuation
WorldModelInstance_desired_state
DefaultModelInstance_actuation
DefaultModelInstance_desired_state
cassie_actuation
cassie_desired_state
geometry_pose
deformable_body_configuration
state
body_poses
body_spatial_velocities
body_spatial_accelerations
generalized_acceleration
net_actuation
reaction_forces
contact_results
WorldModelInstance_state
WorldModelInstance_generalized_acceleration
WorldModelInstance_generalized_contact_forces
WorldModelInstance_net_actuation
DefaultModelInstance_state
DefaultModelInstance_generalized_acceleration
DefaultModelInstance_generalized_contact_forces
DefaultModelInstance_net_actuation
cassie_state
cassie_generalized_acceleration
cassie_generalized_contact_forces
cassie_net_actuation
CassieEncoder
robot_state
filtered_state
Gyroscope
body_poses
measurement
body_spatial_velocities
Accelerometer
body_poses
body_spatial_velocities
body_spatial_accelerations
measurement
SimCassieSensorAggregator
u
x
imu_acceleration
gyro
kcmt_cassie_out
kcmt_radio_out
SceneGraph
drake/multibody/MultibodyPlant@000000012ebc8200_pose
drake/multibody/MultibodyPlant@000000012ebc8200_configuration
query
RgbdSensor
name=pelvis_depth
geometry_query
color_image
depth_image_32f
depth_image_16u
label_image
body_pose_in_world
image_time
MpkOscDiagram
name=osc_controller_for_alip_mpfc
input ports
kcmt_alip_mpc_output
kcmt_robot_output
kcmt_radio_out
RobotOutputReceiver
kcmt_robot_output
x, u, t
AlipMpcOutputReceiver
kcmt_alip_mpc_output
kcmt_fsm_info
footstep_target_in_stance_frame
kx_ky
pitch
u_alip
HighLevelCommand
x, u, t
kcmt_radio_out
pelvis_yaw
pelvis_xy
HeadingTrajGenerator
x, u, t
pelvis_yaw
pelvis_quat
PelvisPitchTrajGenerator
x, u, t
pelvis_pitch
pelvis_quat
SwingToeTrajGenerator
x, u, t
toe_angle
toe_angle
SwingToeTrajGenerator
x, u, t
toe_angle
toe_angle
FsmReceiver
name=FSM Receiver
kcmt_fsm_info
x, u, t
fsm
prev_switch_time
next_switch_time
time_since_prev_switch
time_to_next_switch
CassieAnkleTorqueReceiver
fsm
input_traj
input_value
ud
HipYawTrajGen
fsm
radio
hip_yaw_traj
AlipMPCInterfaceSystem
name=alip_mpc_interface_system
input ports
x, u, t
fsm
touch
swing_ft_traj_interface_system_footstep_target_in_stance_frame
ALIP_com_traj_alip_com_prediction
SwingFootTrajectoryGenerator
name=swing_ft_traj_interface_system
x, u, t
fsm
t_liftoff
t_touchdown
footstep_target_in_stance_frame
swing_foot_xyz_in_stance_frame
output ports
ALIP_com_traj_alip_com_prediction
swing_ft_traj_interface_system_swing_foot_xyz_in_stance_frame
AlipComTrajectoryGenerator
name=ALIP_com_traj
x, u, t
fsm
t1
t0
alip_com_prediction
slope_params
AlipStateCalculator
fsm
x, u, t
alip_state
OperationalSpaceControl
name=OSC
x, u, t
u, t
udes
kcmt_osc_debug
fsm
failure_signal
clock
next_fsm_t_to_impact
lambda_c_des_left_toe
lambda_c_des_left_heel
lambda_c_des_right_toe
lambda_c_des_right_heel
swing_ft_traj
pelvis_balance_traj
alip_com_traj
left_toe_angle_traj
right_toe_angle_traj
swing_hip_yaw_traj
pelvis_heading_traj
RobotCommandSender
u, t
kcmt_robot_input
output ports
u, t
kcmt_robot_input
kcmt_osc_debug
fsm
alip
time_until_switch
LcmPublisherSystem
name=LcmPublisherSystem(ALIP_S2S_MPFC_DEBUG)
channel=ALIP_S2S_MPFC_DEBUG
type=lcmt_alip_mpfc_debug_complete
lcm_message
LcmPublisherSystem
name=LcmPublisherSystem(OSC_WALKING)
channel=OSC_WALKING
type=lcmt_robot_input
lcm_message
LcmPublisherSystem
name=LcmPublisherSystem(OSC_DEBUG_WALKING)
channel=OSC_DEBUG_WALKING
type=lcmt_osc_output
lcm_message
LcmPublisherSystem
name=LcmPublisherSystem(CASSIE_STATE_SIMULATION)
channel=CASSIE_STATE_SIMULATION
type=lcmt_robot_output
lcm_message
drake/examples/cassie/472291936m
PlantVisualizer
input ports
x, u, t
SubvectorPassThrough
u0
y0
MultibodyPositionToGeometryPose
position
geometry_pose
SceneGraph
name=plant_visualizer_scene_graph
Source_279_pose
Source_279_configuration
query
MeshcatVisualizer
name=meshcat_visualizer(visualizer)
path=/drake/visualizer
query_object
Meshcat